



Data Analytics &
Data Engineering

KNIME
Machine Learning

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KNIME Machine Learning Nodes

- Partitioning
- Learner: Algorithm
- Predictor
- Scorer
- Cross-validation

X-Partitioner



X-Aggregator



Naive Bayes Learner



Linear Regression Learner



Decision Tree Learner



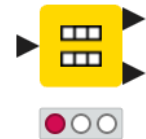
Regression Predictor



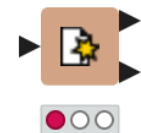
Decision Tree Predictor



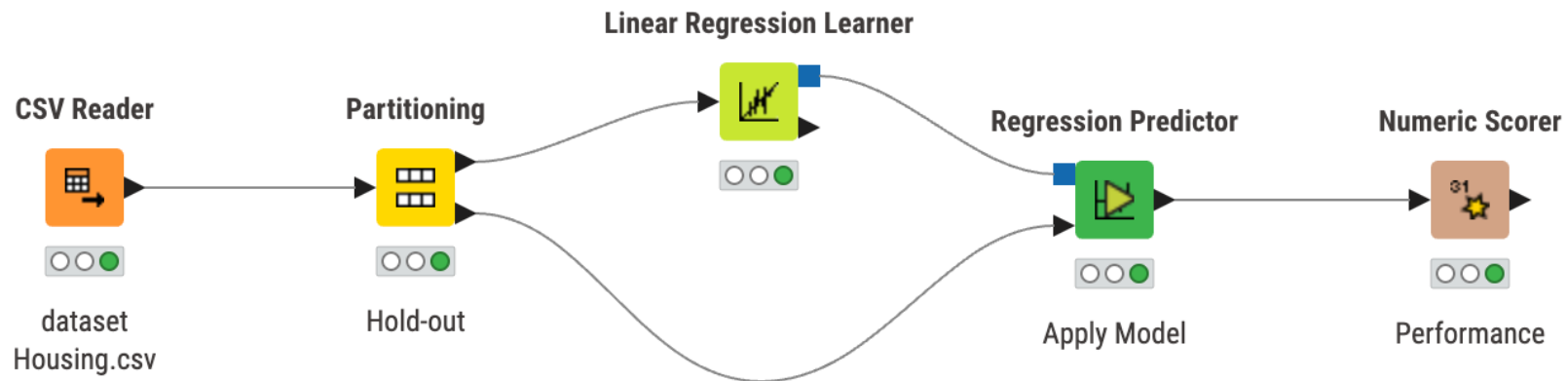
Partitioning



Scorer



Linear Regression



RowID	RM Number (double)	LSTAT Number (double)	PTRATIO Number (double)	MEDV Number (double)
Row0	6.575	4.98	15.3	504,000
Row1	6.421	9.14	17.8	453,600
Row2	7.185	4.03	17.8	728,700
Row3	6.998	2.94	18.7	701,400
Row4	7.147	5.33	18.7	760,200
Row5	6.43	5.21	18.7	602,700
Row6	6.012	12.43	15.2	480,900

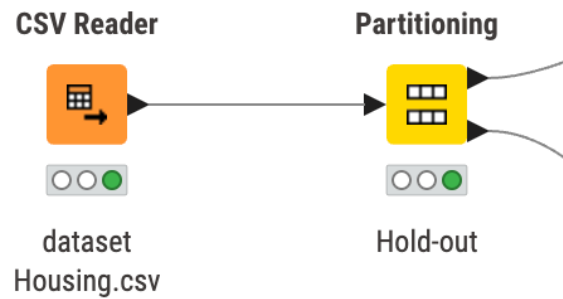
Row No.	RM	LSTAT	PTRATIO	MEDV
1	6.575	4.980	15.300	504000
2	6.421	9.140	17.800	453600
3	7.185	4.030	17.800	728700
4	6.998	2.940	18.700	701400
5	7.147	5.330	18.700	760200
6	6.430	5.210	18.700	602700
7	6.012	12.430	15.200	480900
8	6.172	19.150	15.200	569100
9	5.631	29.930	15.200	346500
10	6.004	17.100	15.200	396900
11	6.377	20.450	15.200	315000
12	6.009	13.270	15.200	396900
13	5.889	15.710	15.200	455700
14	5.949	8.260	21	428400
15	6.096	10.260	21	382200
16	5.834	8.470	21	417900

Table 5.2 Attributes of Boston Housing Dataset

1. CRIM per capita crime rate by town
2. ZN proportion of residential land zoned for lots over 25,000 sq.ft.
3. INDUS proportion of non-retail business acres per town
4. CHAS Charles River dummy variable (= 1 if tract bounds river; 0 otherwise)
5. NOX nitric oxide concentrations (parts per 10 million)
6. RM average number of rooms per dwelling
7. AGE proportion of owner-occupied units built prior to 1940
8. DIS weighted distances to five Boston employment centers
9. RAD index of accessibility to radial highways
10. TAX full-value property-tax rate per \$10,000
11. PTRATIO pupil-teacher ratio by town
12. $B 1000(B_k - 0.63)^2$ where B_k is the proportion of blacks by town
13. LSTAT percent lower status of the population
14. MEDV Median value of owner-occupied homes in \$1000's

ExampleSet (489 examples, 0 special attributes, 4 regular attributes)

Linear Regression



Dialog - 3:4 - Partitioning (Hold-out)

First partition | Flow Variables

Choose size of first partition

- ☐ Absolute 100
- ☒ Relative[%] 70
- ☐ Take from top
- ☐ Linear sampling
- ☒ Draw randomly
- ☐ Stratified sampling
- ☒ Use random seed 1729757668149

OK Apply Cancel ?

Linear Regression



Dialog - 3:2 - Linear Regression Learner

Settings | Flow Variables | Job Manager Selection | Memory Policy

Target: [D] MEDV [v]

Values

☒ Manual Selection ☐ Wildcard/Regex Selection

Exclude

Filter

No columns in this list

☒ Enforce exclusion

Include

Filter

[D] RM
[D] LSTAT
[D] PTRATIO

☐ Enforce inclusion

Regression Properties

☐ Predefined Offset Value: 0 [v]

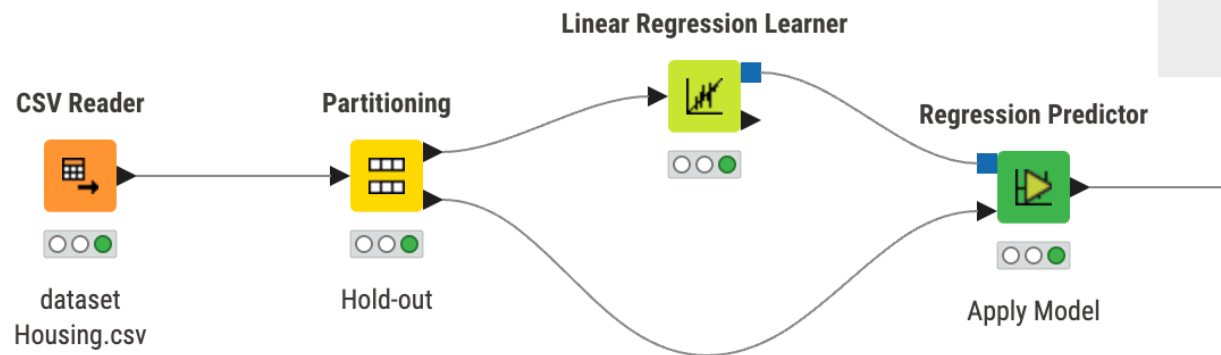
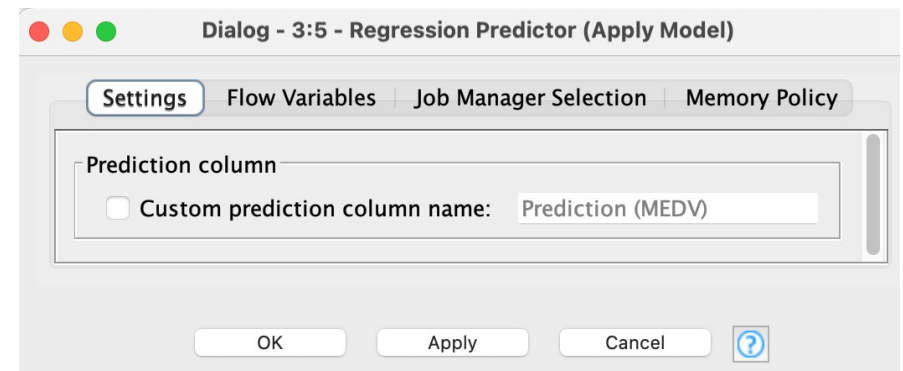
Missing Values in Input Data

☐ Ignore rows with missing values.
☒ Fail on observing missing values.

Scatter Plot View

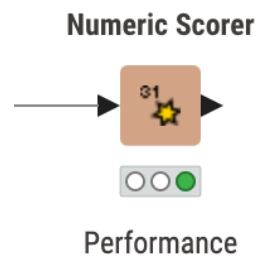
First Row: 1 [v]
Row Count: 20,000 [v]

Linear Regression



RowID	RM <i>Number (double)</i>	LSTAT <i>Number (double)</i>	PTRATIO <i>Number (double)</i>	MEDV <i>Number (double)</i>	Prediction (...) <i>Number (double)</i>
Row2	7.185	4.03	17.8	728,700	644,882.257
Row9	6.004	17.1	15.2	396,900	456,544.309
Row10	6.377	20.45	15.2	315,000	445,468.794
Row13	5.949	8.26	21	428,400	437,605.472
Row15	5.834	8.47	21	417,900	426,517.461
Row18	5.456	11.69	21	424,200	360,552.138
Row20	5.57	21.02	21	285,600	260,242.833
Row22	6.112	18.72	21	310,200	320,042.806

Linear Regression



Dialog - 3:3 - Numeric Scorer (Performance)

Options | Flow Variables | Job Manager Selection | ►

Reference column ▼

Predicted column ▼

Output column

☐ Change column name

Output column name

Provide scores as flow variables

Prefix of flow variables

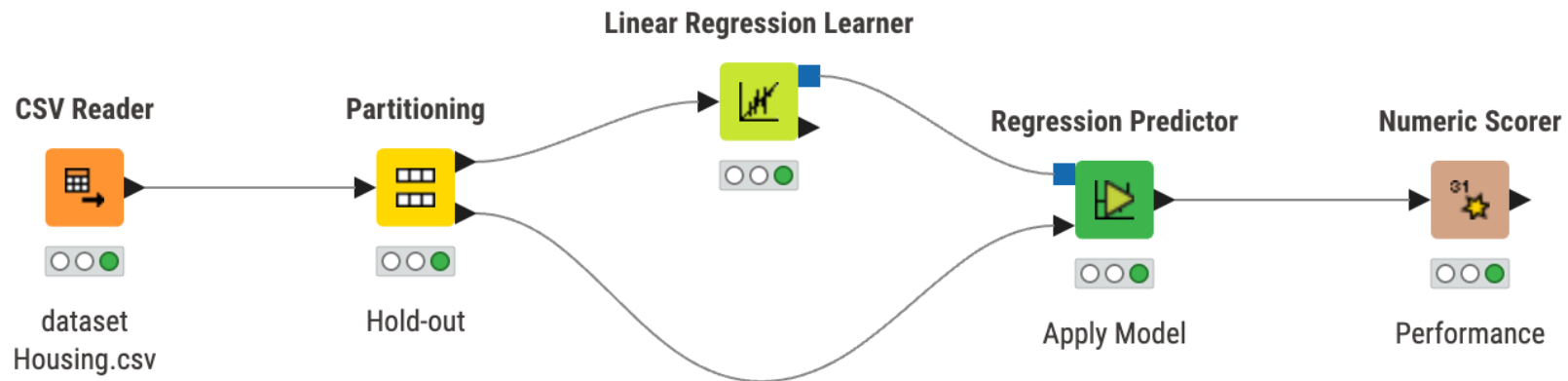
☐ Output scores as flow variables

Adjusted R squared

Number of predictors ▼

OK Apply Cancel ?

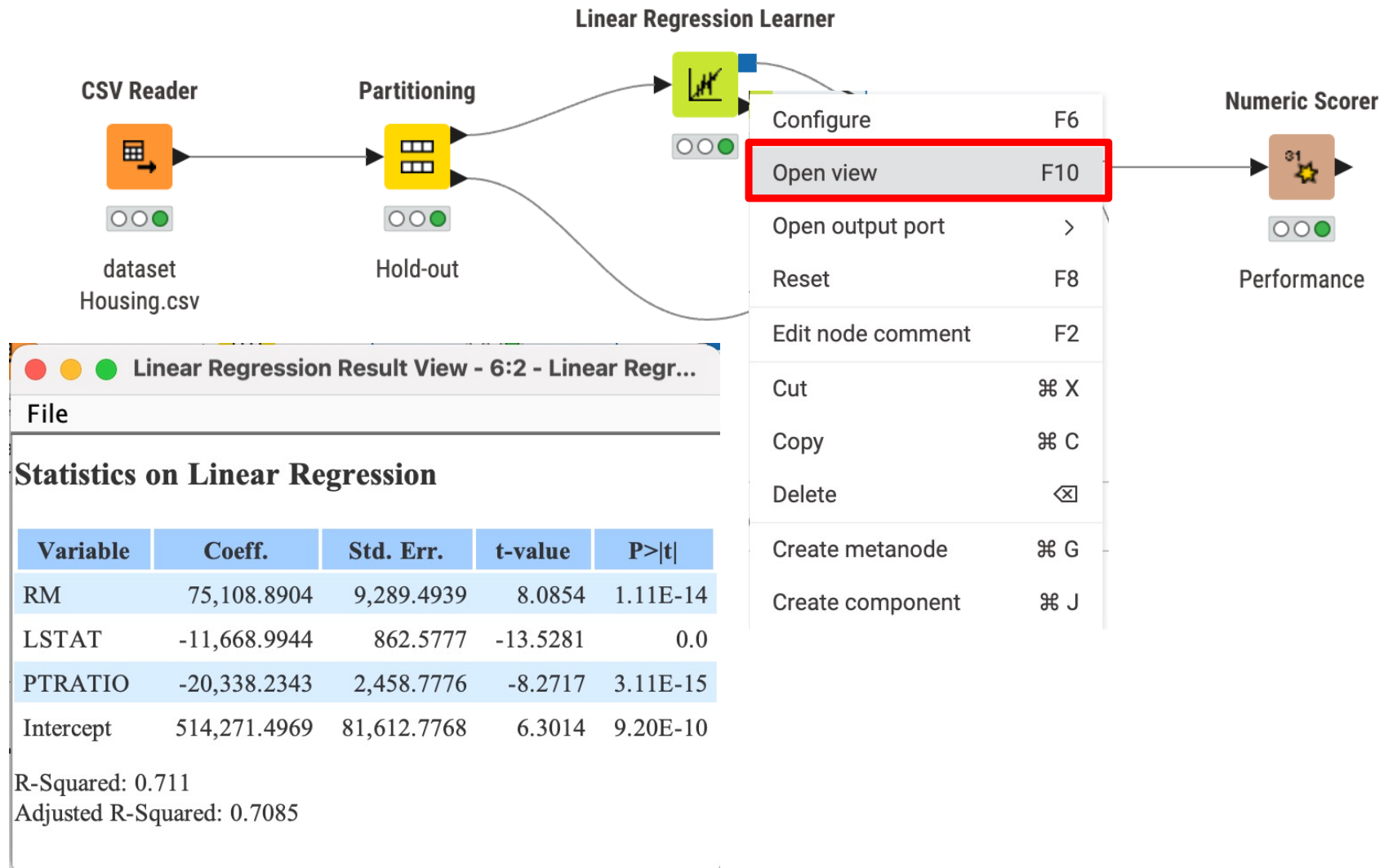
Linear Regression



RowID	RM <i>Number (double)</i>	LSTAT <i>Number (double)</i>	PTRATIO <i>Number (double)</i>	MEDV <i>Number (double)</i>
Row0	6.575	4.98	15.3	504,000
Row1	6.421	9.14	17.8	453,600
Row2	7.185	4.03	17.8	728,700
Row3	6.998	2.94	18.7	701,400
Row4	7.147	5.33	18.7	760,200
Row5	6.43	5.21	18.7	602,700
Row6	6.012	12.43	15.2	480,900

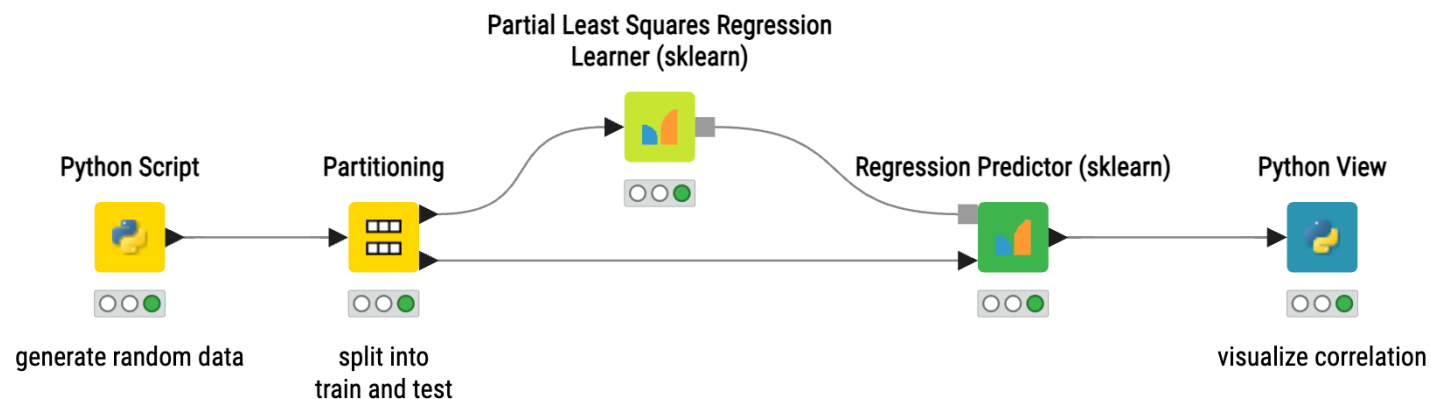
RowID ↑	Prediction (MEDV) <i>Number (double)</i>
R ²	0.727
adjusted R ²	0.727
mean absolute error	65,100.902
mean absolute percentage error	0.163
mean signed difference	3,469.463
mean squared error	7,493,502,670.106
root mean squared error	86,565.02

Right-click



Partial Least Squares Regression Example

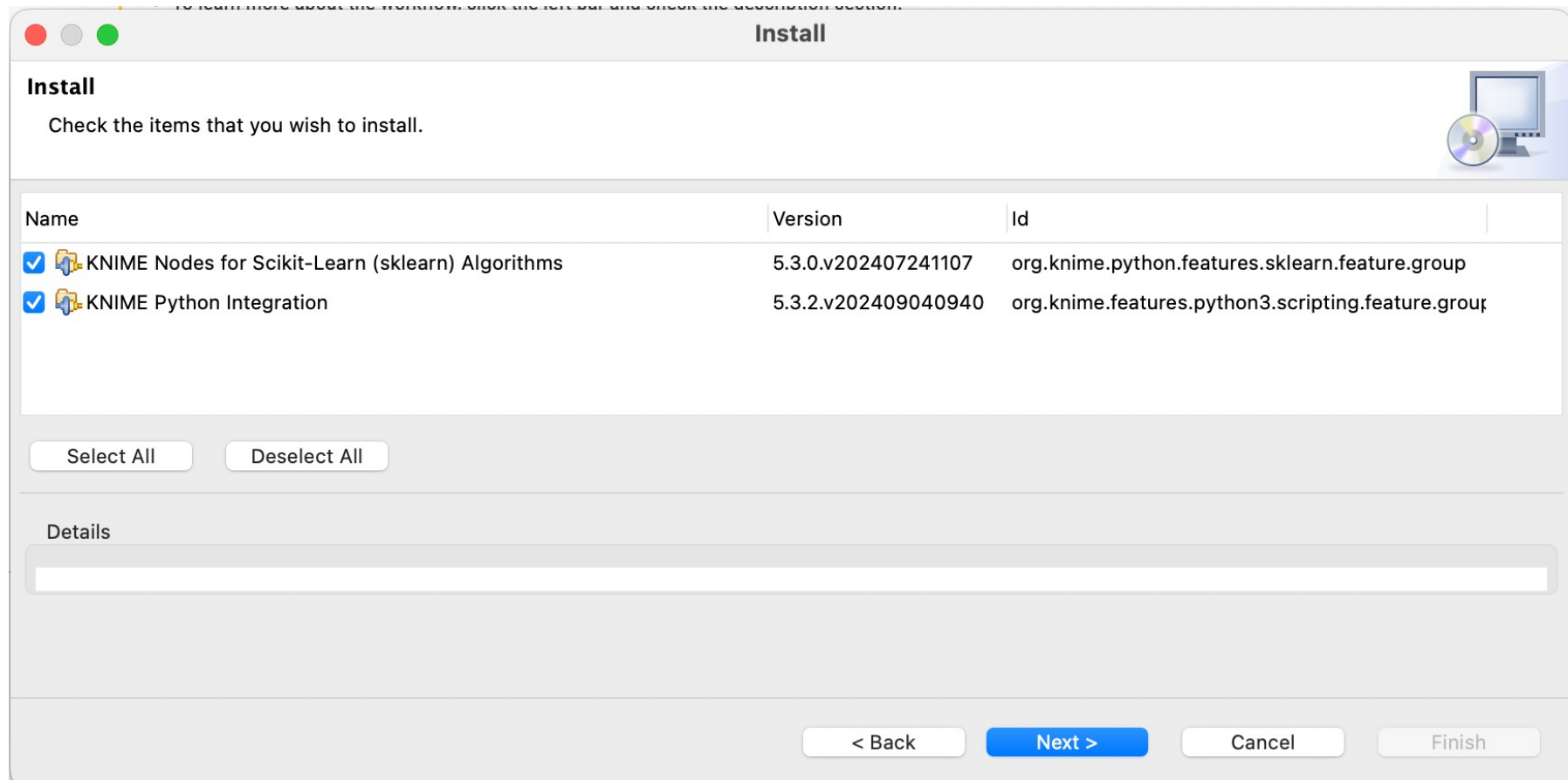
- This example illustrates PLS regression on 2 multivariate covarying two-dimensional datasets.
- Built after the Scikit-Learn example at: https://scikit-learn.org/stable/auto_examples/cross_decomposition/plot_compare_cross_decomposition.html#
- To learn more about the workflow, click the left bar and check the description section.



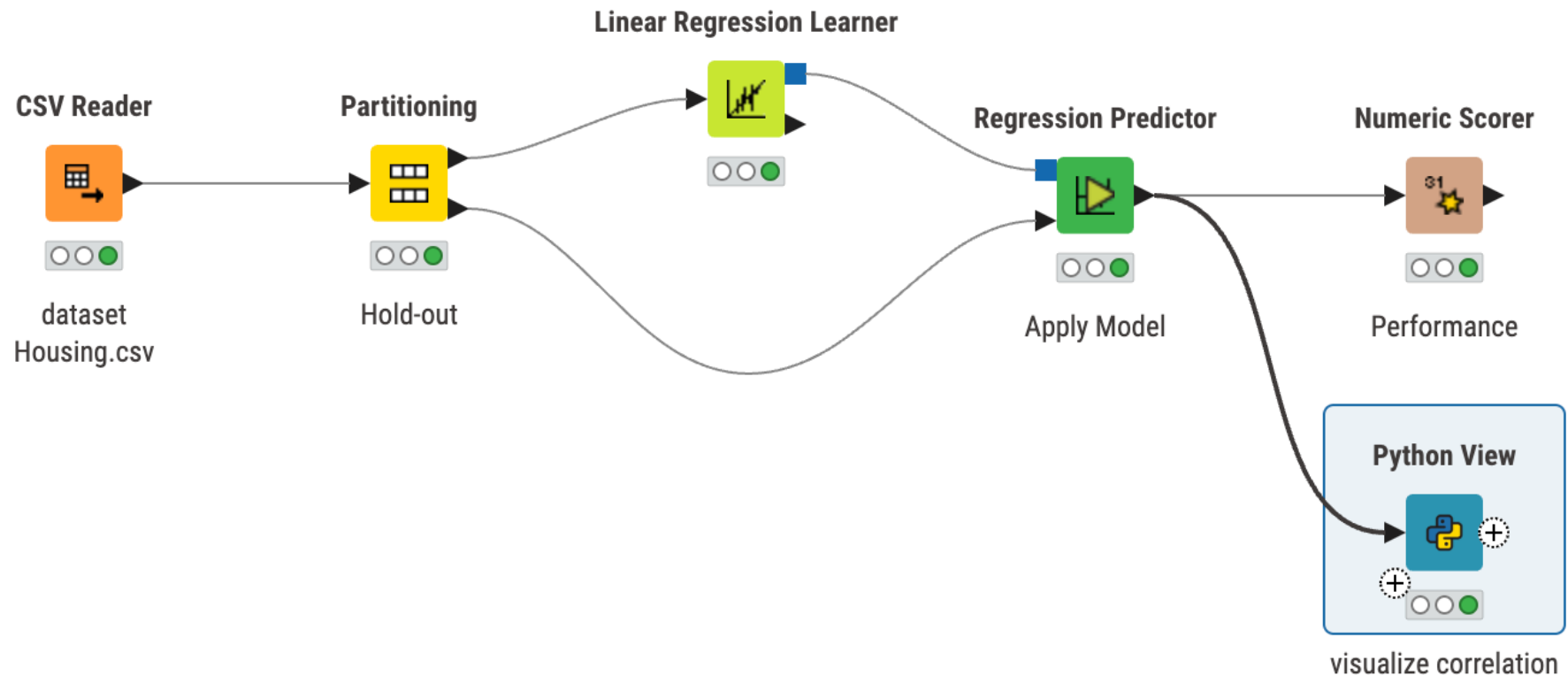
รันและสังเกตผลลัพธ์

https://hub.knime.com/knime/spaces/Examples/04_Analytics/05_Regressions/Partial%20Least%20Squares%20Regression~7o41zTfx3EHYpXpM/current-state

แจ้งเตือน install node และ KNIME Python



ทดลอง Python Script กับ Linear Regression



```
import knime.scripting.io as knio
```

```
import numpy as np
```

```
import matplotlib.pyplot as plt
```

```
# Only use numeric columns
```

```
df = knio.input_tables[0].to_pandas()
```

```
# On diagonal plot X vs Y scores on each components
```

```
plt.figure(figsize=(12, 8))
```

```
plt.subplot(221)
```

```
plt.scatter(df["MEDV"], df["Prediction (MEDV)"], label="prediction", marker="o", s=25)
```

```
plt.xlabel("MEDV")
```

```
plt.ylabel("Prediction (MEDV)")
```

```
plt.title(
```

```
"corr = %.2f"
```

```
% np.corrcoef(df["MEDV"], df["Prediction (MEDV)"])[0, 1]
```

```
)
```

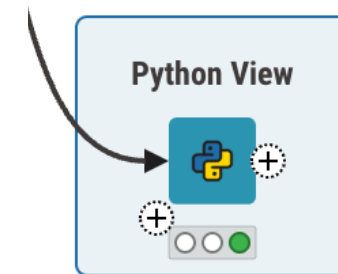
```
plt.xticks(())
```

```
plt.yticks(())
```

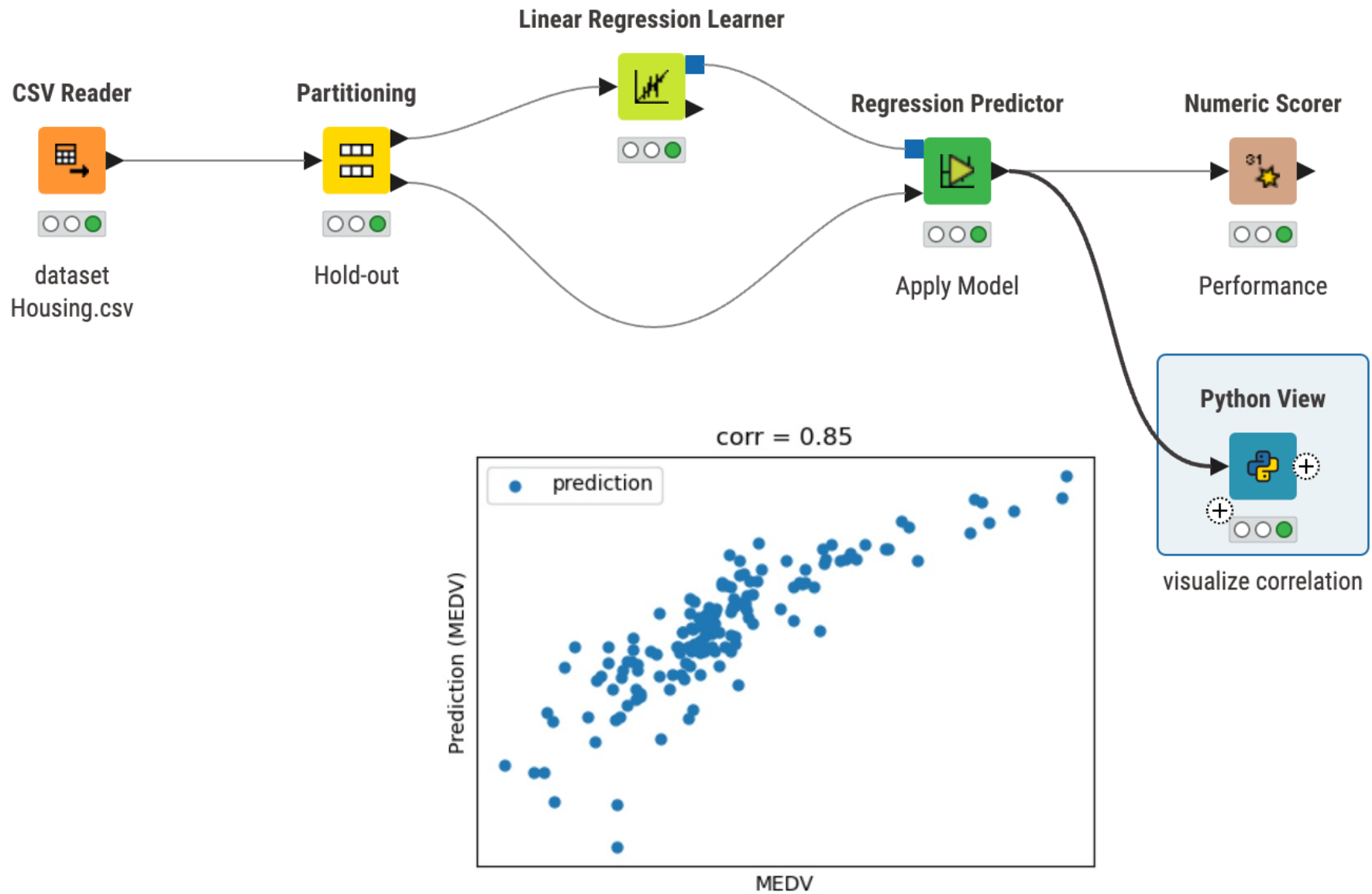
```
plt.legend(loc="best")
```

```
# Assign the figure to the output_view variable
```

```
knio.output_view = knio.view_matplotlib()
```



visualize correlation



Q & A

Cross-Validation

- ใช้ Node สำคัญ 2 ตัว
 - X-Partitioner
 - X-Aggregator

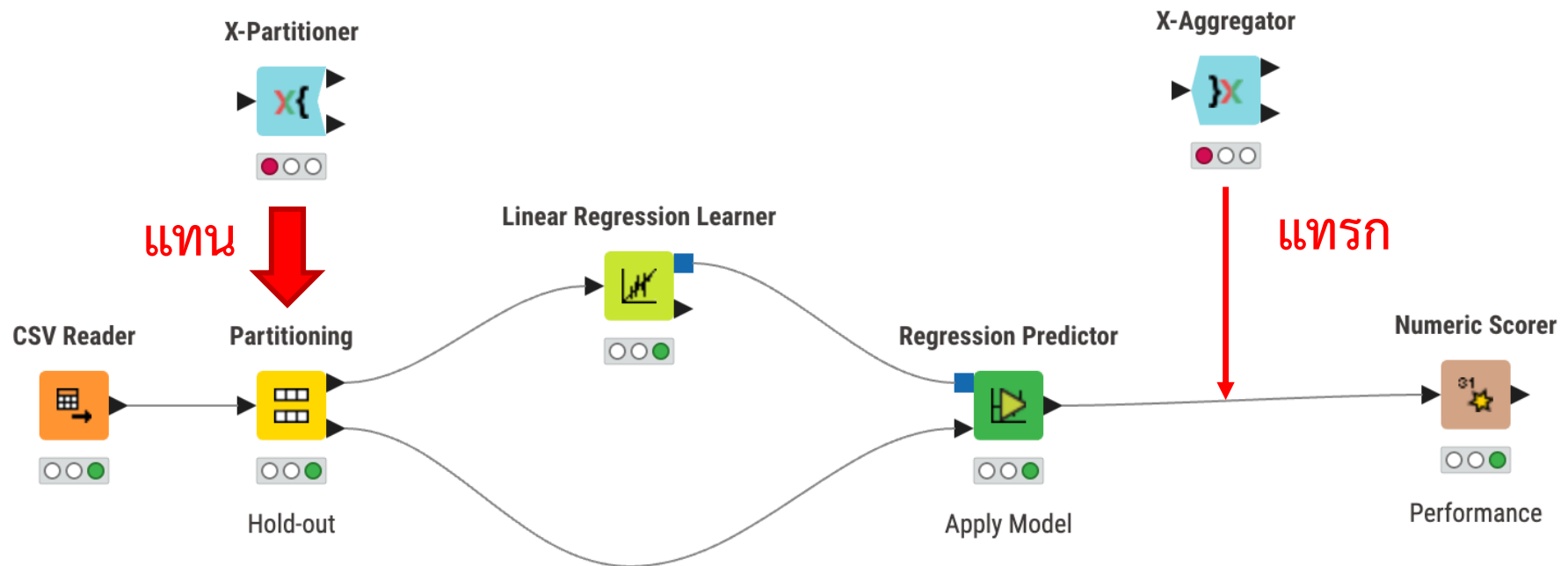
X-Partitioner

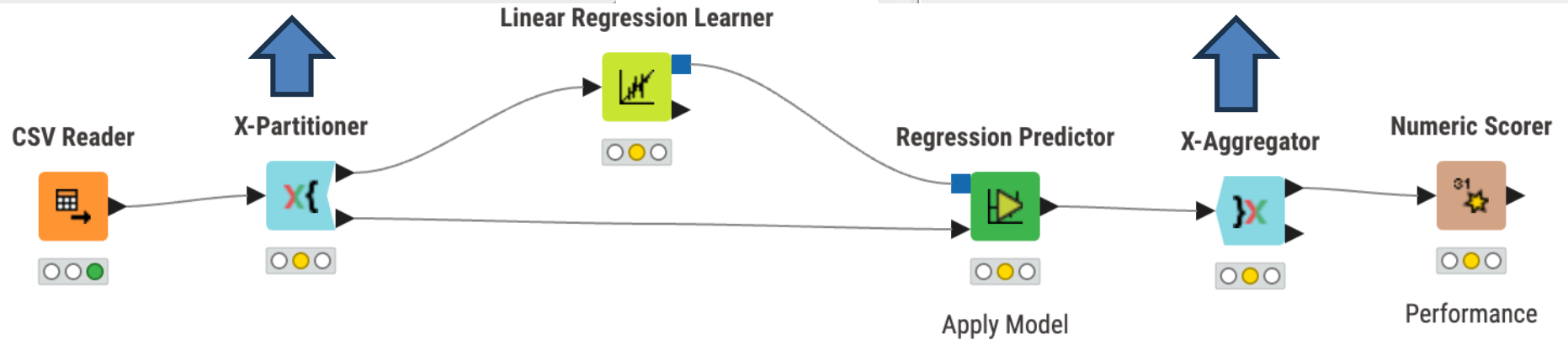
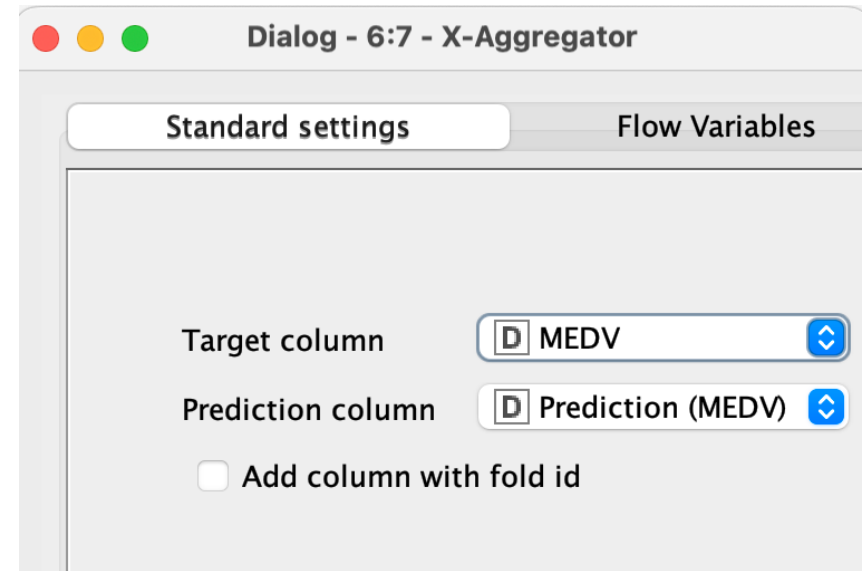
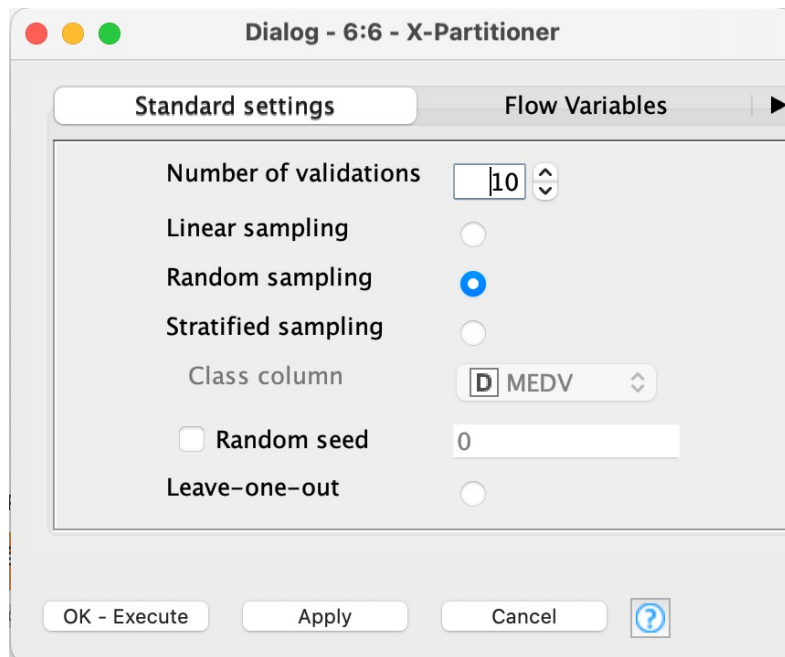


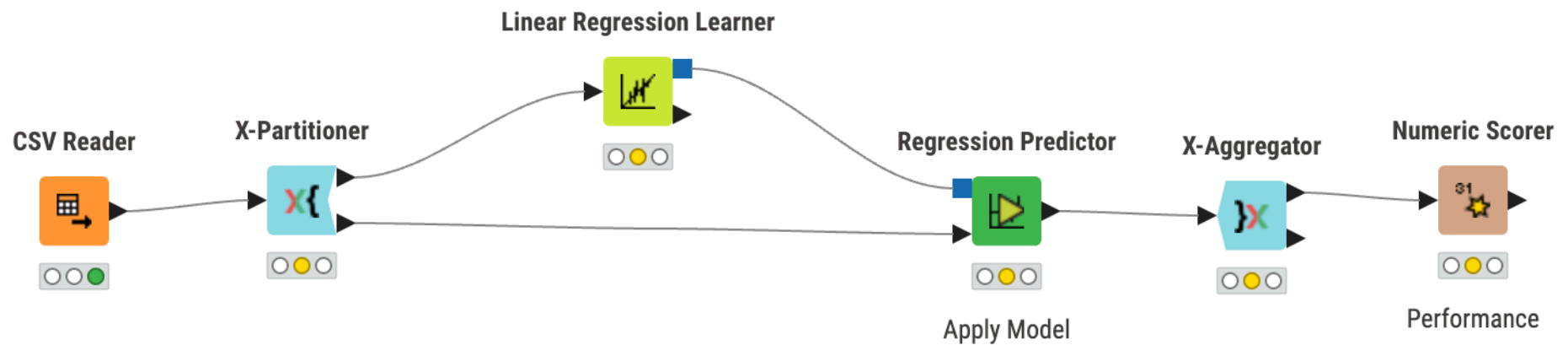
X-Aggregator



Cross-Validation







Hold-out

RowID ↑	Prediction (MEDV) <i>Number (double)</i>
R ²	0.727
adjusted R ²	0.727
mean absolute error	65,100.902
mean absolute percentage error	0.163
mean signed difference	3,469.463
mean squared error	7,493,502,670.106
root mean squared error	86,565.02

10-fold CV

RowID ↑	Prediction (MEDV) <i>Number (double)</i>
R ²	0.71
adjusted R ²	0.71
mean absolute error	66,058.791
mean absolute percentage error	0.174
mean signed difference	-109.116
mean squared error	7,905,499,860.642
root mean squared error	88,912.878

Q & A

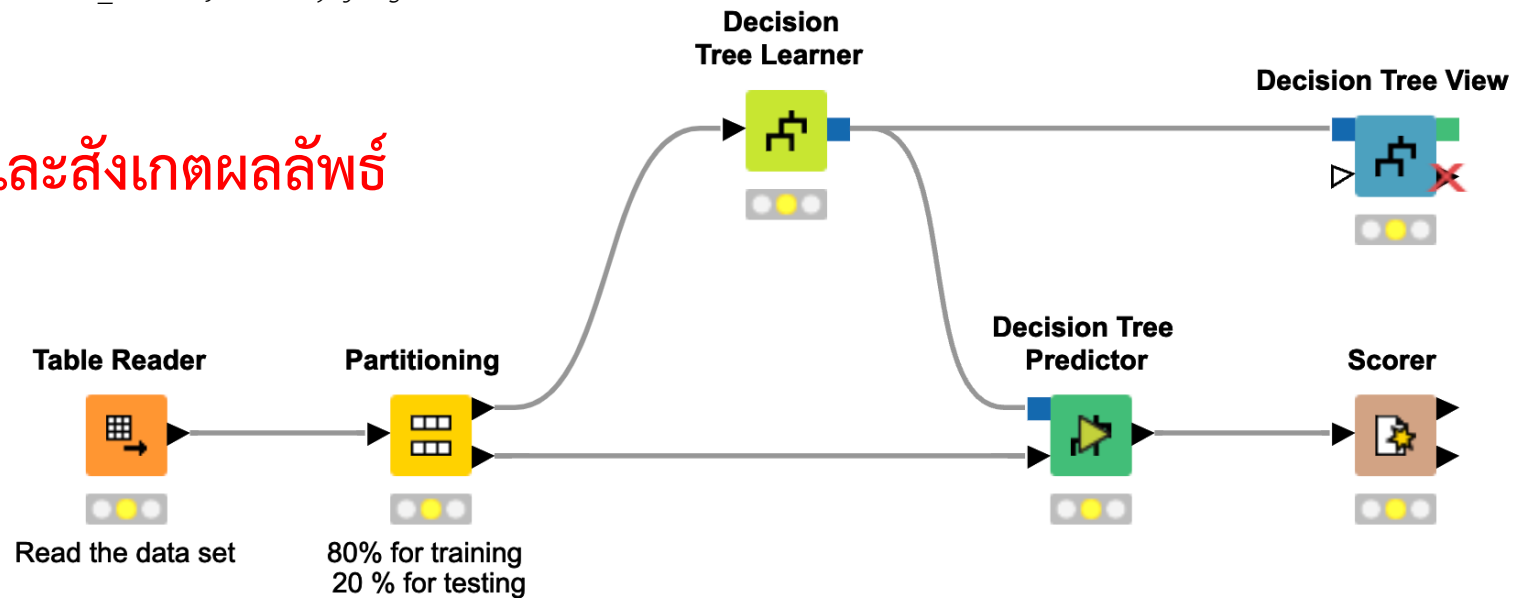
Decision Tree Classification

Decision Tree: Training

This workflow is an example of how to build a basic prediction / classification model using a decision tree. Dataset describes wine chemical features. Output class is wine color: red/white

https://hub.knime.com/knime/spaces/Examples/04_Analytics/04_Classification_and_Predictive_Modelling/07_Decision_Tree~wjBRbBRnly6fu4gw/current-state

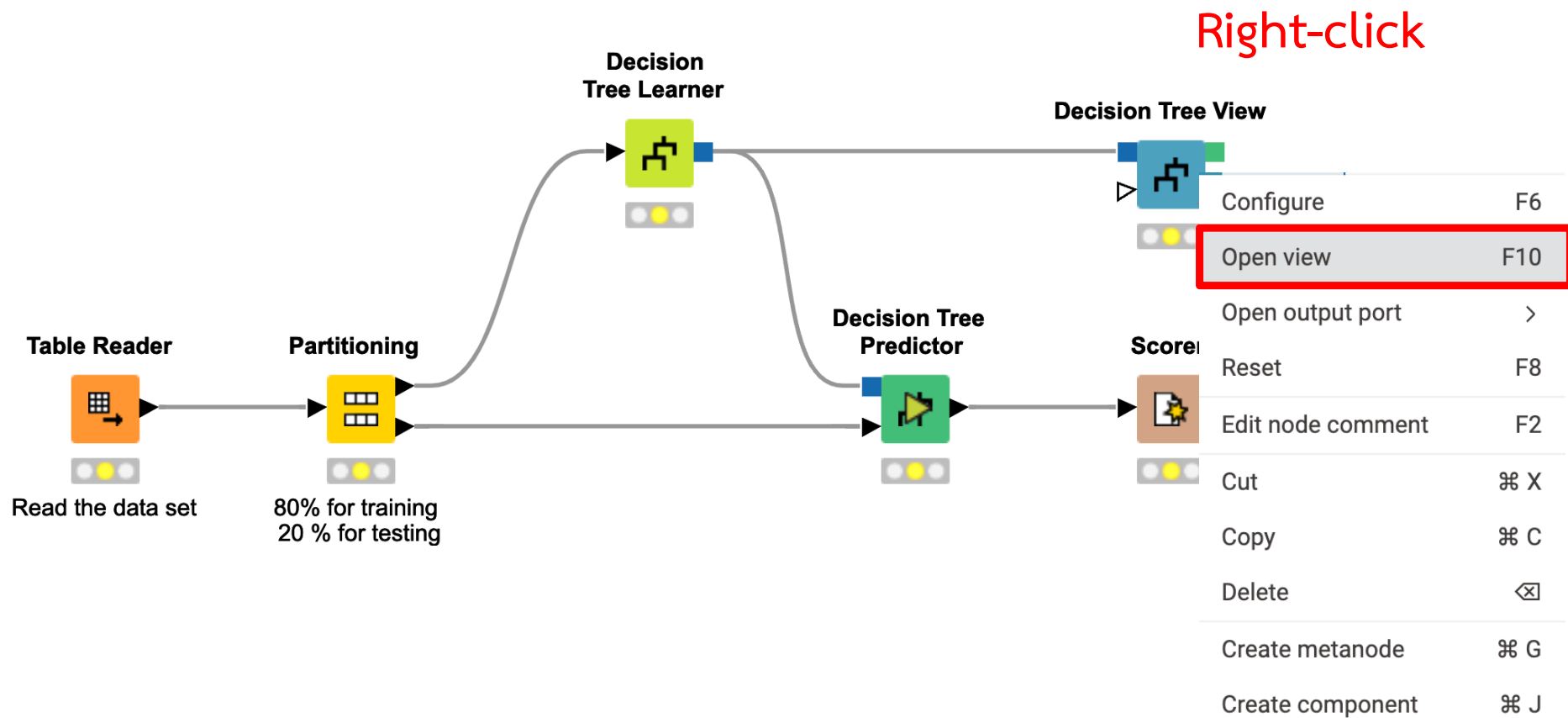
รันและสังเกตผลลัพธ์

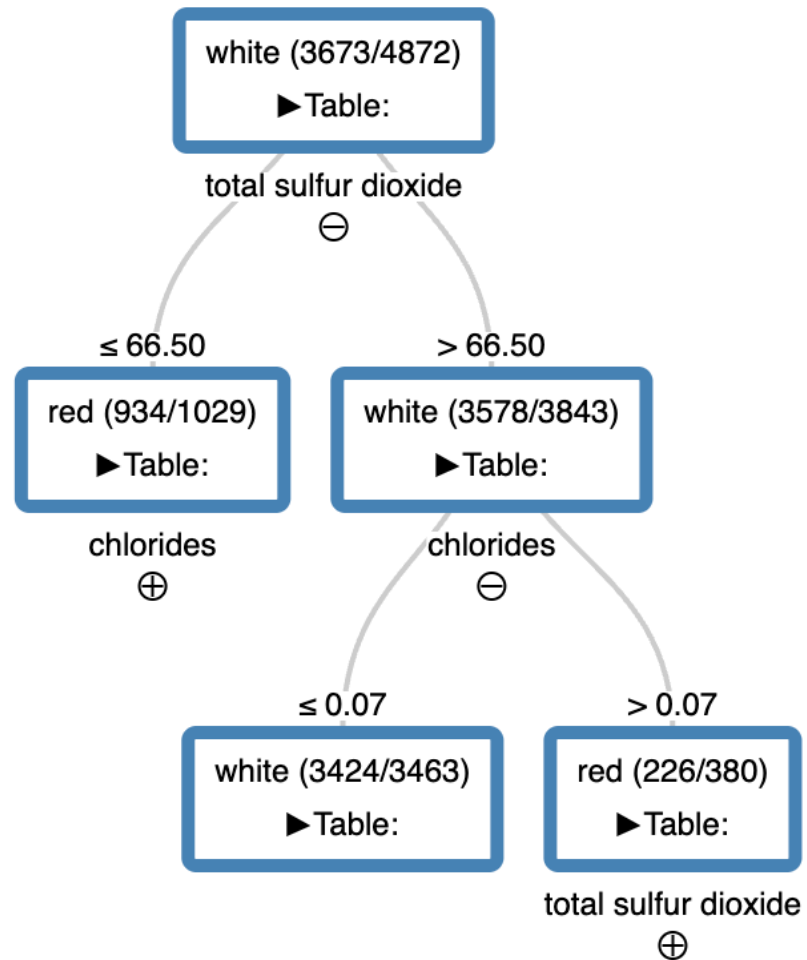


Wine dataset (red/white)

residual s...  <i>Number (dou...</i>	chlorides  <i>Number (dou...</i>	free sulfu...  <i>Number (dou...</i>	total sulf...  <i>Number (dou...</i>	density  <i>Number (dou...</i>	pH  <i>Number (dou...</i>	sulphates  <i>Number (dou...</i>	alcohol  <i>Number (dou...</i>	quality  <i>String</i>	color  <i>String</i>
7.6	0.044	34	144	0.995	3.26	0.38	10.1	neutral	white
4.1	0.201	73.5	189.5	0.994	3.27	0.38	9.4	neutral	white
2.7	0.097	29	65	0.999	3.28	0.58	9.6	slightly dissat...	red
2.2	0.059	11	18	0.992	3.39	0.4	12.8	neutral	red
6.9	0.05	30	97	0.995	3.26	0.44	10.1	neutral	white
1.8	0.036	23	96	0.993	3.32	0.4	10.8	neutral	white
3	0.064	12	27	0.993	3.33	0.59	12.8	neutral	red

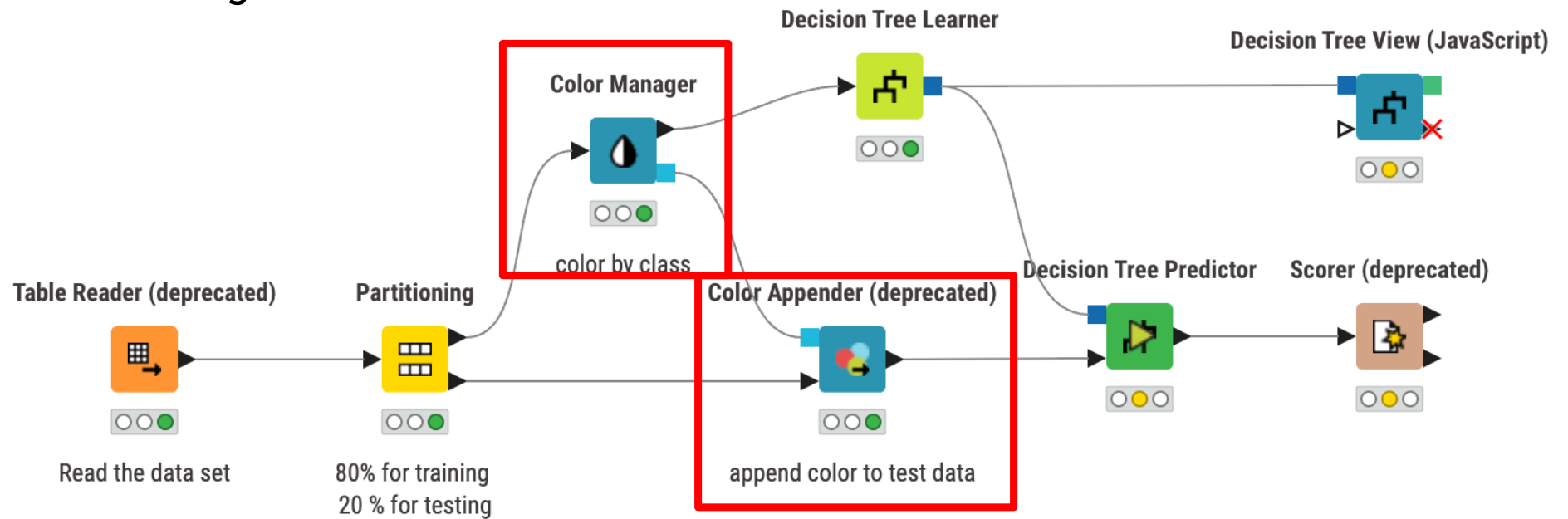
Decision Tree Classification





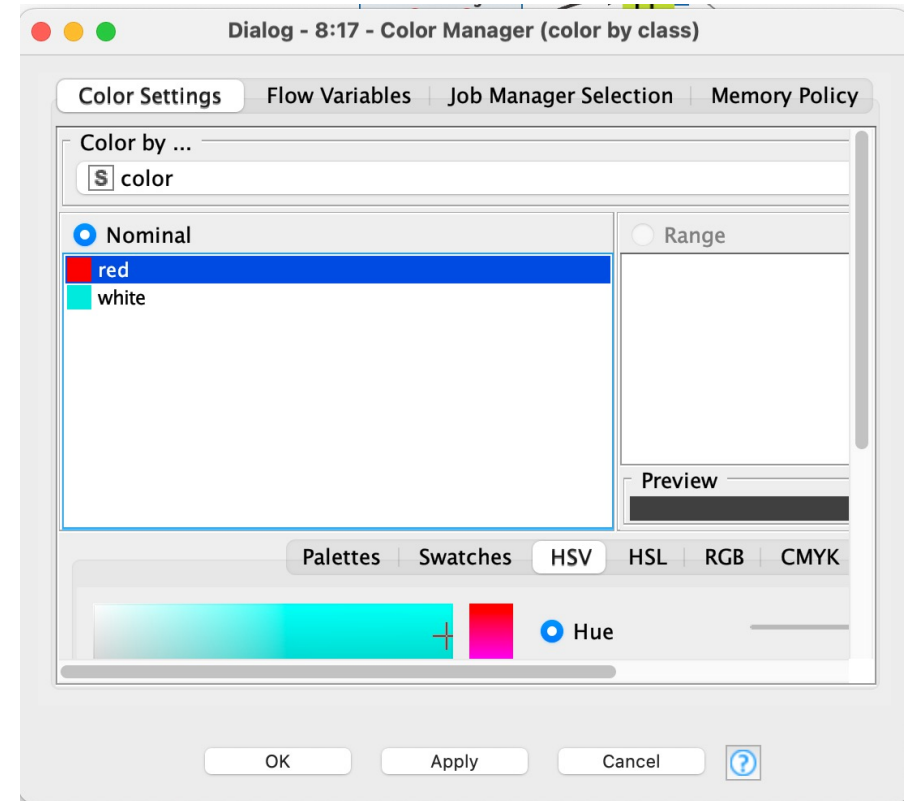
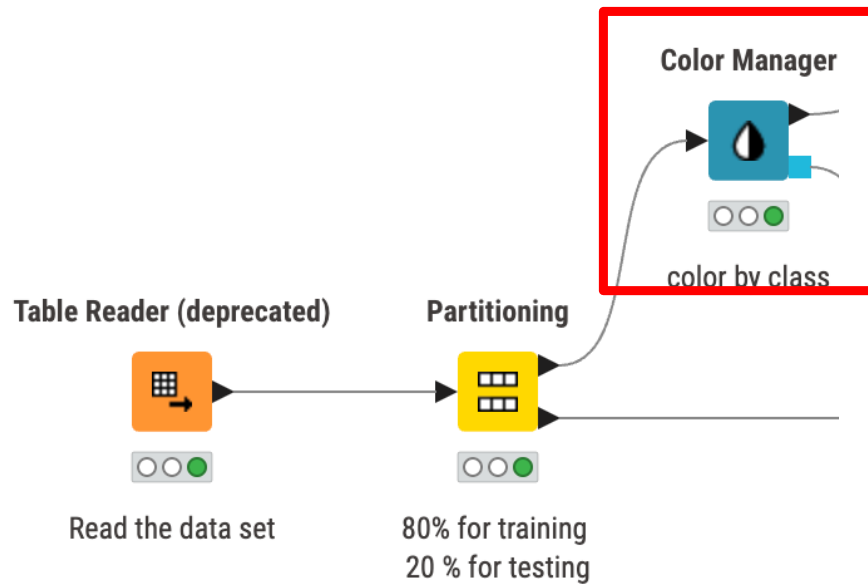
Tip !!!

Color Manager



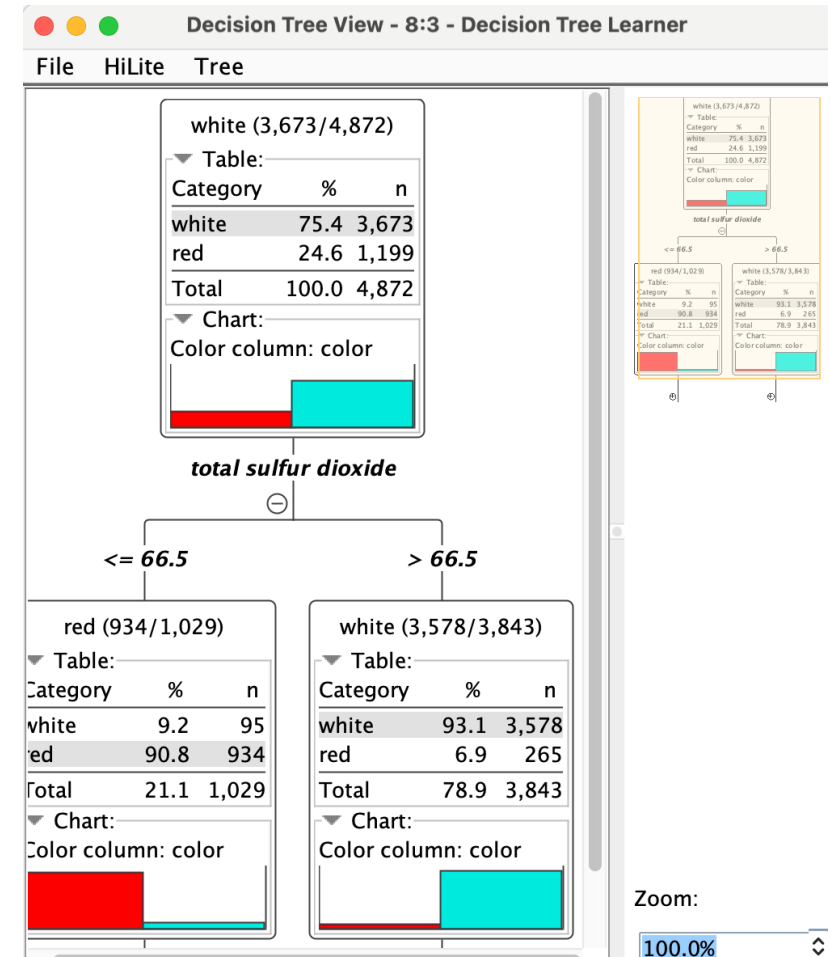
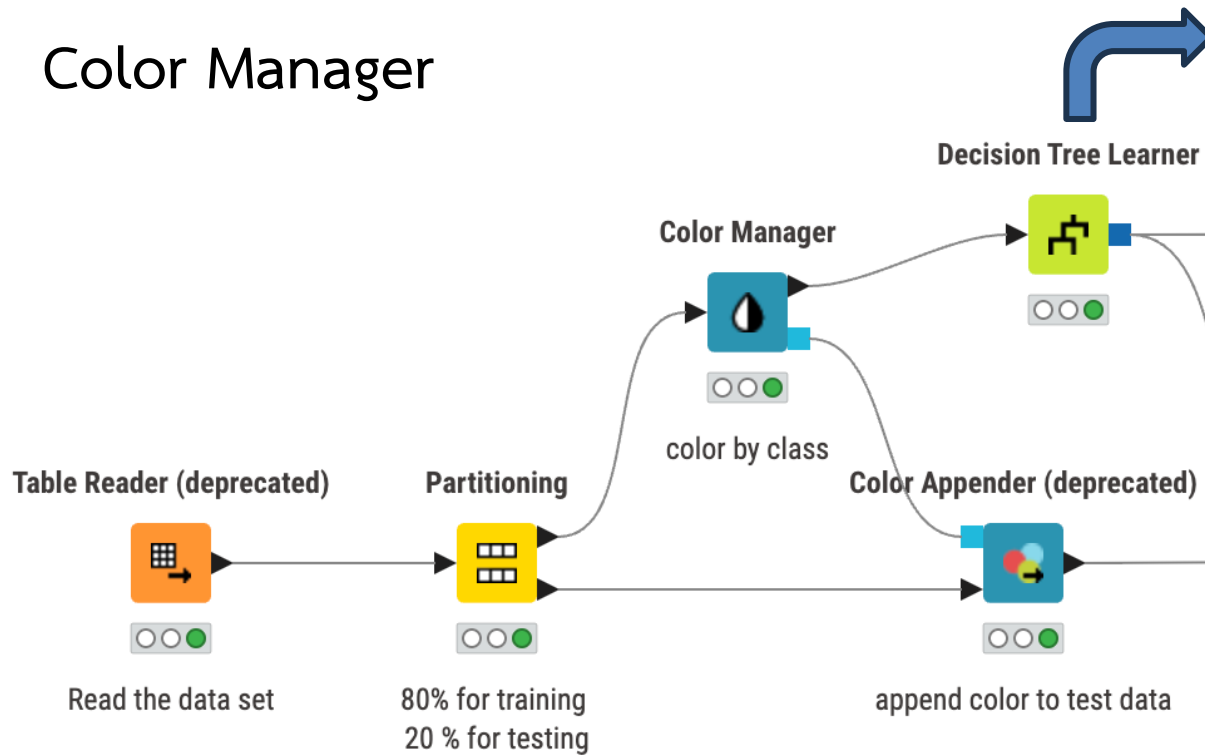
Tip !!!

Color Manager

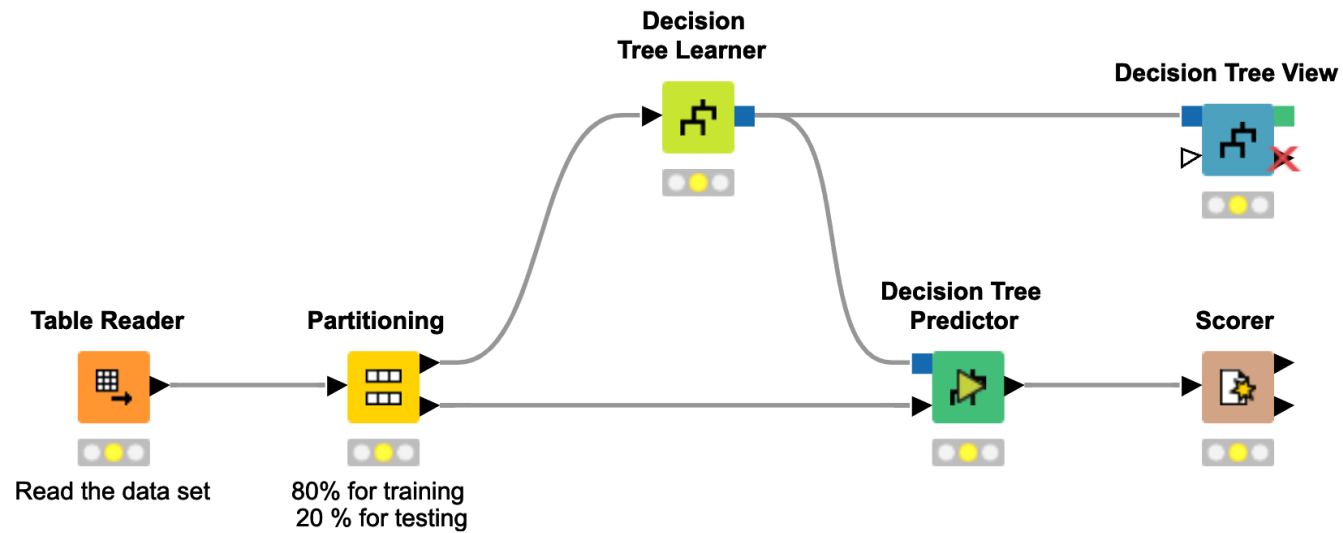


Tip !!!

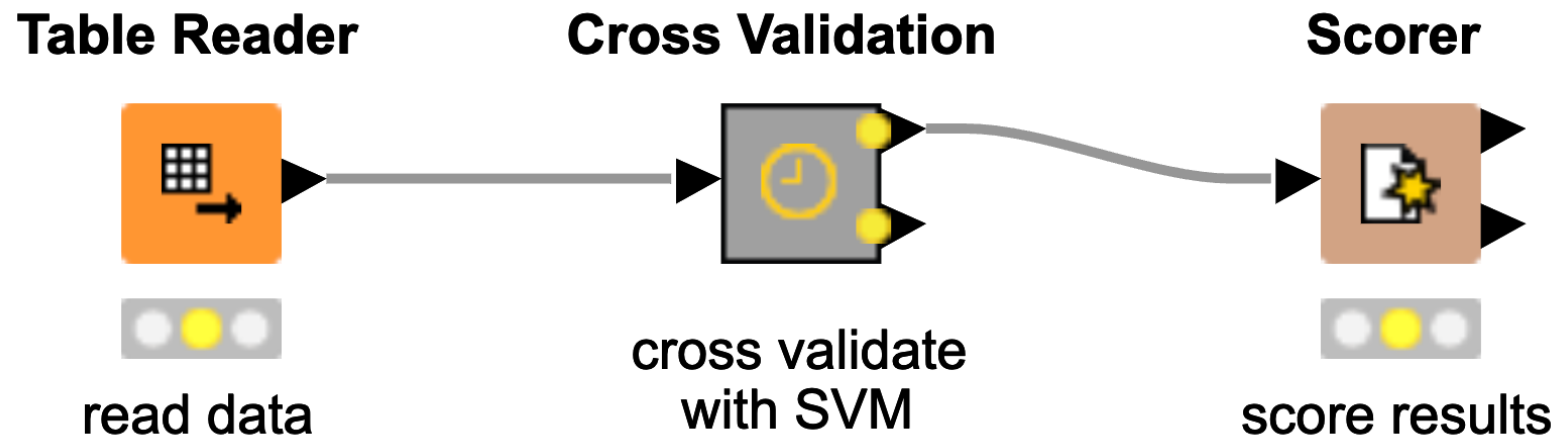
Color Manager



LAB: ทดลอง 10-fold cross validation



ตัวอย่างการใช้ Metanode



https://hub.knime.com/knime/spaces/Examples/04_Analytics/11_Optimization/01_Cross_Validation_with_SVM~JWHivBU7q_pL-bF9/current-state

Q & A